

SUDBURY, ON, Canada

WMO: 717300

Lat: 46.6237N

Lon: 80.7939W

Elev: 1143

StdP: 14.10

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-12.6	-27.7	1.2	-17.9	-22.1	1.7	-11.5	25.5	17.1	23.1	17.5	7.0	330	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	84.2	68.1	81.1	66.2	78.2	64.6	70.8	79.9	68.9	76.8	67.2	74.4	11.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.0	107.2	74.2	66.2	100.7	72.8	64.5	94.9	71.1	35.4	80.1	33.8	76.6	32.3	74.5	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.5	19.9	17.9	-24.9	89.5	5.9	3.4	-29.1	91.9	-32.6	93.9	-35.9	95.8	-40.2	98.2		
(5)			-25.3	74.5	5.8	1.9	-29.4	75.9	-32.8	77.0	-36.1	78.1	-40.3	79.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.1	9.6	12.8	23.7	37.8	52.5	62.3	66.4	64.4	56.9	44.0	30.6	18.1		
	DBStd	22.02	13.52	11.40	11.50	9.22	8.49	7.03	5.47	5.66	7.76	8.46	9.81	11.67		
	HDD50	5390	1254	1041	816	380	72	4	0	0	24	229	582	988		
	HDD65	9328	1719	1461	1279	817	397	131	49	79	260	652	1031	1453		
	CDD50	1767	0	0	1	13	148	373	509	448	232	42	1	0		
	CDD65	233	0	0	0	0	9	50	93	62	18	1	0	0		
	CDH74	1782	0	0	0	2	108	428	711	417	113	3	0	0		
	CDH80	377	0	0	0	0	21	94	164	81	17	0	0	0		
	WSAvg	8.8	9.2	9.4	9.5	9.9	9.1	7.9	7.3	7.5	8.2	9.1	9.3	9.0		
	PrecAvg	36.1	2.5	2.1	2.6	2.7	3.0	3.5	3.1	3.5	4.2	3.0	3.2	2.8		
(15) Precipitation	PrecMax	44.1	5.0	4.6	5.0	7.1	5.7	7.7	7.9	7.9	9.6	6.0	6.9	4.1		
	PrecMin	26.4	0.7	0.6	1.0	0.7	0.9	1.5	1.1	0.9	1.0	0.4	0.6	1.2		
	PrecStd	4.5	1.0	1.1	1.2	1.3	1.3	1.7	1.7	1.7	2.0	1.1	1.3	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.8	40.9	56.8	70.7	83.1	86.8	88.9	86.5	82.2	71.8	56.7	44.8		
		MCWB	40.0	37.5	47.9	54.4	64.4	68.8	71.0	70.2	68.5	60.7	50.8	42.9		
	2%	DB	35.7	36.1	48.6	63.0	76.6	82.8	84.5	82.0	77.2	65.7	51.5	39.3		
		MCWB	34.3	34.0	41.8	48.9	59.9	66.5	68.9	67.2	65.7	57.7	48.6	37.9		
	5%	DB	32.3	33.1	43.3	57.0	72.1	79.0	80.9	78.6	73.1	61.1	47.5	36.2		
		MCWB	31.3	31.3	37.9	45.9	57.4	64.2	66.3	65.2	63.1	55.4	45.1	35.0		
	10%	DB	29.6	30.1	39.0	52.1	67.5	75.4	77.9	75.4	69.3	56.7	44.1	33.6		
		MCWB	28.3	28.2	34.4	42.8	55.1	62.1	64.9	63.8	61.4	52.5	41.9	32.5		
	0.4%	WB	40.3	38.8	50.7	57.5	67.7	72.4	74.1	72.3	70.7	63.2	53.1	43.4		
		MCDB	40.8	40.3	52.9	64.1	78.0	84.5	84.5	81.2	79.0	67.2	55.5	44.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.6	34.5	43.0	51.5	63.5	69.2	71.1	69.8	67.6	59.4	49.0	38.3		
		MCDB	35.4	35.9	46.9	58.6	72.0	78.4	80.1	77.2	73.2	64.0	50.8	39.1		
	5%	WB	31.5	31.4	38.5	47.2	60.2	66.9	69.0	67.9	65.5	56.1	45.5	35.1		
		MCDB	32.3	32.8	42.7	55.4	67.7	75.2	76.9	74.9	70.7	60.3	47.3	36.0		
	10%	WB	28.4	28.3	34.8	43.4	56.8	64.5	67.0	66.0	63.1	53.0	41.8	32.5		
		MCDB	29.6	30.0	38.3	50.8	65.1	71.7	74.5	72.4	68.6	56.5	43.9	33.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	17.1	17.5	17.8	19.9	19.1	19.0	18.0	17.4	14.0	12.0	12.8		
		MCDBR	14.1	15.3	19.7	25.7	25.8	23.3	23.1	21.9	21.5	20.0	16.6	12.3		
		MCWBR	14.0	14.1	14.0	15.4	13.2	11.0	10.6	9.9	11.3	13.0	14.2	11.8		
		MCDBR	14.0	14.7	18.0	23.6	22.4	19.7	19.6	18.2	18.3	18.0	16.0	12.3		
	5% WB	MCWBR	14.1	14.1	13.9	15.9	13.1	11.0	10.7	10.1	11.4	13.2	14.7	12.2		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.282	0.302	0.343	0.362	0.387	0.389	0.380	0.357	0.325	0.298	0.264		
	taud		2.312	2.243	2.315	2.366	2.391	2.343	2.362	2.381	2.441	2.517	2.511	2.406		
	Ebn at Noon		265	282	293	290	286	279	276	274	272	266	253	254		
	Edh at Noon		23	31	34	36	36	39	37	35	30	24	20	19		
(44) All-Sky Solar Radiation	RadAvg		403	724	1177	1454	1699	1810	1820	1529	1180	704	387	302		
	RadStd		59	90	108	125	150	142	122	100	116	87	46	27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page